

Krrish Punj

[LinkedIn](#) | [Email](#) | [Github](#)

EDUCATION

- **Thapar Institute of Engineering and Technology** Patiala, India
Bachelor of Engineering in Computer Sciences 2023 – 2027
Relevant Coursework: Conversational AI, Numerical Linear Algebra, Operating Systems, Computer Architecture CGPA: 8.00

EXPERIENCE

- **Undergraduate Research Intern** Dec 2025 – May 2026
CODSAI (in collaboration with Imperial College, London) On-Site | Patiala, India
 - Designed CNN-based computational kernels for FDM fluid simulations on grids exceeding 100+M points, on H100 DGX node infrastructure.
 - Developed solver pipelines using the **PETSc** library for large sparse linear systems on high resolution meshes via **GMRES** and **CG** solvers preconditioned with **AIR**, **ILU** and **multigrid** methods.
 - Extended the **PFLARE** library by adding new solver test cases and verifying functional correctness across various parameter ranges
- **Software Development Intern** June 2026 – July 2026
LG Electronics On-Site | Noida, India
 - Built an AI driven manufacturing analytics platform for realtime line monitoring, anomaly detection, incident tracking across plant operations.
 - Implemented a RAG based workflow using pgvector, Ollama, and local LLM inference to process query over live and historical production data and generate context aware responses.
 - Integrated backend APIs, PostgreSQL persistence, realtime socket updates, and Docker based deployment to support data-driven decision support for plant operations.

SKILLS

Languages: C++(CUDA), Python, Julia, MATLAB, JavaScript, SQL, Bash/Shell Scripting
Frameworks & libraries: PyTorch, Tensorflow, PETsc, Nodejs, Reactjs, Nextjs, OpenMP
Databases: PostgreSQL, Oracle, Mongodb
DevOps & Cloud: Docker, Git, Linux Internals

PROJECTS

- **Euler Easel** | *C++, CUDA, OpenMP, AVX, Python* **GitHub**
 - Built a hardware-aware adaptive runtime for Sparse Matrix Vector Multiplication (SpMV) that analyzes matrix characteristics, execution state, and hardware capabilities to select efficient CPU and GPU execution strategies.
 - Implemented high performance CSR, ELLPACK, and Hybrid (CSR-ELL) sparse formats with custom **CUDA** kernels and optimized CPU backends using **OpenMP** and **AVX** intrinsics.
 - It is a hierarchical execution planner that choose the most suitable execution strategy for the current matrix and workload using one-step reinforcement learning.
- **Feels Like Summer** | *Next.js, Go, PostgreSQL, Docker, GORM* **GitHub**
 - Built a university research collaboration platform enabling students to discover research opportunities and apply to projects through a centralized portal.
 - Implemented RESTful backend services in Go with GORM, backed by a normalized PostgreSQL schema for project management and application workflows.
 - Optimized backend scalability using **15 worker goroutine concurrency**, thread safe **5 minute TTL** in-memory caching, batch database queries to eliminate N+1 lookups, and indexed PostgreSQL tables for efficient data retrieval.

PUBLICATIONS

- Author, **Krrish Punj**, Author, Smile Bansal, **Physics Informed Neural Network Modeling and Uncertainty Quantification of Heat Transfer in Casson Fluid under Local Thermal Non-Equilibrium Effects**, Indian Journal of Physics, 2026. (Manuscript Ref. No. INJP-D-26-00306R2)
- Author, **Krrish Punj**, Author, Dr. Amrita Kaur, **SAHARA: Articulatory Dynamics-Guided Silent Speech Recognition Using Physics-Informed Neural Networks for Punjabi Vocabulary**, Computer Methods and Programs in Biomedicine, 2026. (communicated)